… To Operations Management in Manufacturing and Service Industries

Transforming resources, production methods, site and capacity, purchasing and inventory (JIT, MRP), Gantt and PERT charts, services (Burger King), total quality management, ISO and outsourcing.

01 The common thread: producing the PowerSki Jetboard

Bob Montgomery spent **sixteen years** designing the Jetboard, with thousands of design changes. The product won *Time* magazine's "**Best Invention of the Year**" award. But having a product was only the beginning: the company needed a **system that would produce high-quality Jetboards at reasonable prices**.

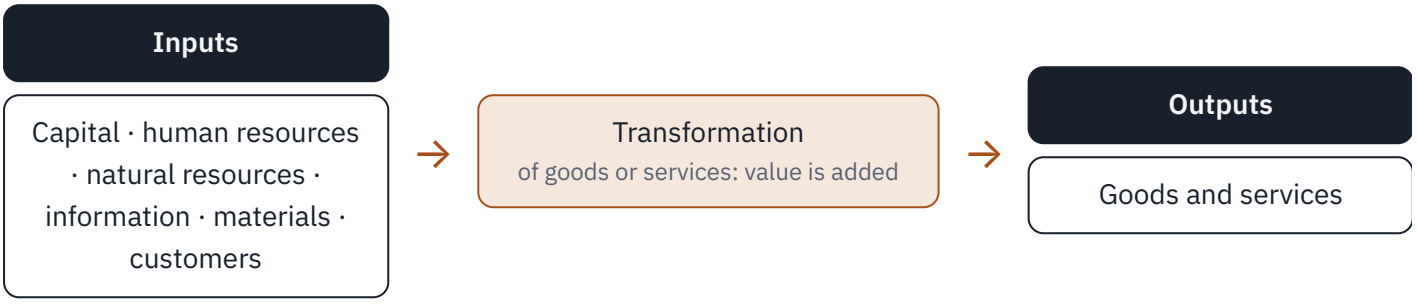
POWERSKI'S PRODUCTION QUESTIONS

- What kind of **production process** should it use?
- How **large** should the facilities be, and **where**?
- How should the plant be **laid out**?
- Make every component in-house or use **subcontractors**?
- Where to buy the **materials**?
- Which systems ensure **efficiency** and **quality**?

The chapter answers these questions for manufacturing, then for services (Burger King), and ends with quality and outsourcing.

02 Operations management

Operations management (OM): management of the process that **transforms resources (inputs) into products (outputs)**. The person in charge is the **operations manager**.



To compete, a company must convert resources **as efficiently as possible**: focus on quality, hold down the costs of materials and labor, and eliminate all costs that add **no value** to the finished product.

The operations manager's three responsibilities

RESPONSIBILITY	CONTENT
Production planning	**How** goods will be produced (process), **where** (site selection), and **how facilities will be laid out**
Production control	Schedule and monitor activities, respond to feedback and adjust; oversee purchasing of raw materials and handling of inventories
Quality control	Ensure goods are produced to specifications and quality standards are maintained

TRADE-OFFS

Production choices depend on the goals set by marketing: low cost, high end, reliability, a wide range of products... They involve trade-offs: low cost doesn't normally go hand in hand with high quality.

03 Production methods, site and capacity

Three production methods

METHOD	PRINCIPLE	EXAMPLES
Make-to-order strategy	Products made to **customer specifications**: low volume, high variety	Print and sign shops; furniture and clothing made by craftspeople
Mass production (make-to-stock strategy)	**High volumes** of identical goods at **low cost**, made in anticipation of demand (forecasts) and held **in inventory**	Processed foods, electronic appliances, cars
Mass customization	Fairly high volumes of **customized** products at fairly low prices: mass-produce up to a cut-off point, then customize	Nike iD (Nike By You), Levi's jeans by curve type (*slight, demi, bold, supreme*), Oakley, M&M's in school colors

REMEMBER

Mass customization doesn't work for every type of good: most people don't care about customized detergents or paper products, and many aren't willing to pay the higher prices customization commands.

Site selection

- Locate **close to suppliers, customers, or both** (shipping costs).

- Areas with ample numbers of **skilled workers**.
- Locations where managers and their families will **enjoy living**.
- **Low costs** for land, labor, construction, utilities and taxes.
- A **favorable business climate** (e.g. tax breaks).

POWERSKI IN SAN CLEMENTE, CALIFORNIA

Three priority criteria: **proximity to suppliers**, **availability of skilled engineers and technicians**, **favorable living conditions**. Low costs and local incentives mattered less; proximity to customers was unimportant (worldwide distribution).

Capacity

Capacity: the **maximum number of products** that a facility can produce over a given period **under normal working conditions**.

You first forecast demand (Chapter 7), then calculate capacity requirements, the investment needed and labor hours. **Capacity too low**: lost sales and customers. **Capacity too high**: wasted resources and inflated operating costs.

04 Facility layouts NOT FOR THE MCQ

NOT FOR THE MCQ

Your course material marks section 8.3 “**NOT FOR THE MCQ**”.

LAYOUT	PRINCIPLE	EXAMPLE
Process layout	Groups together workers or departments that perform similar tasks, with specialized equipment	Vermont Teddy Bear (cutting, sewing, stuffing…)
Product layout (assembly line)	High-volume goods produced in assembly-line fashion: a series of workstations where already-made parts are assembled	Peeps marshmallows at Just Born
Cellular layout	Small teams handle all aspects of building a component, a family of components or a finished product (often U-shaped)	Boeing, Raytheon, Pratt & Whitney, Little Enterprise
Fixed-position layout	Large items stay in one place; workers and equipment go to the product	Ships, buildings; General Housing homes (Bay City, Michigan)

05 Managing the production process: purchasing, inventory, scheduling

Materials management: covers **purchasing**, **inventory control** and **work scheduling**.

Purchasing

Material costs often make up **about 50%** of total manufacturing costs, so vendor selection (**purchasing / procurement**) gets a lot of attention. Questions: can the supplier provide the right **quantity** at the right **price**? Its **quality**? Is it **reliable** (on-time delivery)? Does it have a good **reputation**? Is it easy to work with?

- **E-purchasing (e-procurement)**: companies use the internet to interact with suppliers (online catalogs, e-marketplaces, bidding), like a consumer buying a TV online. Lower costs, faster procurement, better communication.
- **Electronic data interchange (EDI)**: computerized exchange of business transaction documents.

Inventory control

Inventory control: managing inventory so the company has **enough to keep operations flowing smoothly**, **but not so much** that money is wasted holding it (storage, insurance).

METHOD	PRINCIPLE
Just-in-time (JIT) production	Suppliers deliver materials **just in time** to enter the production process: little inventory, but a lot of communication and cooperation with suppliers
Material requirements planning (MRP)	A computer program calculates the **quantity of materials** needed and **when** to order them. The book's analogy: a fund-raising dinner for 50 guests (menu → ingredients → schedule → shopping)
Manufacturing resource planning (MRP II)	Coordinates MRP with the activities of the firm's **other functional areas** (e.g. HR labor forecasts)

Work scheduling

Master production schedule (MPS): a timetable that specifies **which and how many products will be produced and when**. It relies on **routing**: the workflow of each item based on the sequence of operations in which it will be used.

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06 Gantt and PERT charts

Gantt chart (Henry Gantt): an easy-to-use graphical tool for determining the **status of projects**. Useful when the production process is fairly simple and activities aren't interrelated.

100 "HIKING BEARS" AT VERMONT TEDDY BEAR (FIGURE 8.9)

By **day 6**, all accessories and clothing have been made. The **stuffing and sewing** of the fur (days 3 to 8) isn't scheduled for completion until the end of day 8: this is the activity to watch, because it must be finished before the bears are dressed (days 9 to 11), packaged (day 12) and shipped (day 13).

PERT chart (Program Evaluation and Review Technique): a tool for diagramming the activities required to produce a good, specifying the **time** each activity takes, and organizing them in the **most efficient sequence**. It identifies the **critical path**: the **sequence of activities that will entail the greatest amount of time**.

ONE HIKING BEAR (FIGURE 8.10)

- Fur: cut 10 min → stuff and sew 30 min = **40 min**
- Clothes: cut 10 → sew 10 → embroider T-shirt 10 = 30 min
- Accessories (backpack, tent): cut 5 → sew 10 = 15 min
- Then, together: dress 15 → package 5 → ship 5 = 25 min

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Critical path: cut fur → stuff and sew → dress → package → ship = 65 min
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WHY IT MATTERS

To produce a bear faster, you must save time **on the critical path**. Speeding up the clothes achieves nothing: they would just wait for the fur at the dressing station.

07 The technology of goods production NOT FOR THE MCQ

NOT FOR THE MCQ

Your course material marks section 8.6 "**NOT FOR THE MCQ**".

TOOL	ROLE
CAD (computer-aided design)	Creates models representing the design of a product (PowerSki sends design changes to the suppliers of its 340 components)
CAM (computer-aided manufacturing)	Determines the production steps and instructs machines
CIM (computer-integrated manufacturing)	Integrates design, production and functional activities, from order taking to shipping
FMS (flexible manufacturing system)	Computer-controlled equipment that can easily be adapted to produce a variety of goods

- **Automation**: replacing workers with machines.

- **Robots**: advanced automation carrying out complex series of actions; autonomous or semi-autonomous (Flippy the burger-flipping robot, delivery robots).
- **3D printing** (additive manufacturing): building an object layer by layer from a CAD model. Cheaper than traditional manufacturing for prototypes, labor, material waste and inventory; suited to low volumes and complex, small or customized parts.

08 Operations management for service providers

The U.S. economy has become a service economy: only about **13%** of workers are in the goods-producing sector (manufacturing 8%, construction 4%, mining 1%) and **80%** in services. The largest employers are Walmart, Amazon, UPS, Yum! Brands, Kroger and Home Depot; only the seventh, IBM, has a manufacturing component.

Three differences from manufacturing

INTANGIBILITY
Banking, entertainment and education can't be touched or handled.

CUSTOMIZATION
One bottle of Pepsi is like any other; a haircut is tailored to the customer.

CUSTOMER CONTACT
Many services are bought and consumed at the same time.

The Burger King case

More than 19,000 restaurants in the U.S. and about a hundred other countries; around 11 million customers a day.

- **Process**: BK uses a **make-to-order** approach (sandwiches built one at a time: a special order takes 30 seconds); McDonald's uses **make-to-stock** (batches of identical sandwiches: a burger without onions can take up to five minutes).
- **Efficiency**: BK was the first to offer **drive-through** service (80% of sales); David Sell moved the **drink station** in front of the counter (one fewer employee per day per restaurant, lower costs) and replaced boxes with a single **paper wrapper**.
- **Site**: "*through the light and to the right*"; the key criterion is the **traffic count**, the number of cars or people passing a location each day.
- **Size**: a prime acre went from $100,000 to over $1 million; the split moved from 90% inside dining / 10% drive-through to 80 / 20; restaurants shrank from 4,000 to as little as 1,000 square feet, down to 10-by-10-foot kiosks.

CAPACITY AND SCHEDULING IN SERVICES

Services **can't be stored** (hairdressers can't inventory haircuts): capacity must meet needs on an **as-demanded basis**. In manufacturing, managers schedule **activities**; in services, they schedule **workers** to handle fluctuating demand. BK uses data from its **point-of-sale** registers for staffing and inventory.

09 Producing for quality

Quality (American Society for Quality): "the **characteristics of a product or service** that bear on its ability to **satisfy stated or implied needs**".

Total quality management (TQM): all the steps a company takes to ensure its products satisfy customer needs. Three principles:

1. CUSTOMER SATISFACTION
Customers define quality; satisfaction surveys are tracked over time.

2. EMPLOYEE INVOLVEMENT
Everyone is responsible; **quality circles**: employees doing similar jobs who identify problems and propose solutions.

3. CONTINUOUS IMPROVEMENT
A commitment to constant improvement in design, production and delivery.

Tools

- **Statistical process control (SPC)**: monitoring quality by testing **samples** of output against specifications. Example: Kellogg's wants **two scoops of raisins** in every box of Raisin Bran; boxes are pulled off the line and recorded on a **control chart**; if several samples in a row are low, the line is shut down.
- **Benchmarking**: comparing your performance with that of a company that excels in the same activity. Frequent targets: **L.L. Bean** (filling orders), **3M** (innovation), **Motorola** (consistent quality), **Mary Kay** (inventory control).

International standards

STANDARD	AREA
ISO 9000	**Quality** management
ISO 14000	**Environmental** management

Set by the International Organization for Standardization (ISO, a nongovernmental agency based in Switzerland). They focus on **the way a company does its work**, not on its output; compliance is **voluntary** and certification is time-consuming and complex, but almost essential to compete in the global marketplace.

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10 Outsourcing

Outsourcing: using **outside vendors** to manufacture all or part of a company's products (or to provide services).

POWERSKI
After its long design period (ending in May 2003), PowerSki had outside vendors make the engines, fiberglass hulls and parts, and final assembly took place at **Experience Powersports** in **Moses Lake**, Washington. Every component is made to PowerSki's exact specifications and inspected on arrival: the company **keeps control of quality** and focuses on design.

Why outsource?

- Few companies have the **expertise** to produce everything: firms specialize in what they do best.
- Take advantage of **lower labor costs** in other regions.

- In **services** too: schools often outsource food service, the bookstore or IT; companies outsource IT support or call centers.

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11 The chapter's cases

- **Ghost restaurant**: a delivery-only restaurant with no dining room. Less risky (no dining room or servers, lower rent), but dependent on delivery platforms.
- **McDonald's**: the McDonald brothers streamlined the kitchen like a production line; **Ray Kroc** grew the chain through franchising.
- **Ethics**: BP's **Deepwater Horizon** disaster, where obvious safety hazards were ignored to meet a cost-based deadline.
- **Nike iD / By You**: customizing shoes online or in studios; a customizable billboard in Times Square.
- **Outsourcing and jobs**: should employees worry about job security when a company outsources?

Revision sheet based on the textbook Exploring Business (FlatWorld, version 5.0.3), chapters 7 and 8. Sections marked "not for the MCQ" are labeled "NOT FOR THE MCQ" in the course material; worked answers are suggestions.